# Min Jae Bae, Ph.D.

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## Education

- Ph.D. in Biomedical Science and Engineering, Konkuk University, 2021 – Present (expected Feb. 2027) — Advisor: Prof. Dae-Jin Yun
- M.S. in Biomedical Science and Engineering, Konkuk University, 2019 – 2021 — Thesis: Structural and Functional Analysis of HOS15, a Substrate Receptor of CRL4 in Cold Stress Signaling
- B.S. in Biomedical Science and Engineering, Konkuk University, 2013 – 2019 — Thesis: Functional Studies of SUMOylation of Histone Deacetylase in Plants

## Research Focus

My research investigates how the HOS15–DDB1–CUL4 E3 ubiquitin ligase complex regulates chromatin structure and gene expression during cold stress in *Arabidopsis thaliana*, focusing on its functional interplay with the histone deacetylase HD2C and the linker histone H1.3. I study how HOS15-mediated changes in histone acetylation and methylation reshape chromatin states, and how this translates into stress-responsive transcriptional reprogramming.

At the protein level, I dissect the structure–function relationship of HOS15 through site-directed, single-amino-acid mutagenesis of its LisH motif to map residues required for its activity and protein conformation, and I characterize the intrinsically disordered region (IDR) within its coiled-coil domain. To connect these molecular changes to global stress responses, I use LC-MS-based proteomics to profile proteins that accumulate in HOS15 mutants following environmental stress treatment, integrating domain-level biochemistry with gene expression and proteome-wide analysis.

## Research Projects

- Global Plant Stress Research Center (NRF), 2024.08–2030.04 — Participating Researcher
- Genome-wide chromatin reprogramming and transcriptome regulation in plant environmental stress adaptation (NRF), 2026.03–2031.02 — Participating Researcher
- Chromatin reconfiguration and transcriptional control for plant stress adaptation (NRF), 2022.03–2027.02 — Co-participating Researcher

## Professional Service

- Member, NASA GeneLab OSDR Plant Analysis Working Group (Plant AWG), Sep. 2026–Present — Contributing to the analysis of spaceflight and astrobotany omics datasets within NASA's Open Science Data Repository (OSDR), collaborating with international researchers on plant biology under the space environment.
- Member, NASA GeneLab OSDR Regolith Analysis Working Group (Regolith AWG), Sep. 2026–Present — Participating in cross-disciplinary efforts on plant and plant–microbe growth in lunar/Martian regolith, contributing plant molecular biology expertise to regolith-related dataset curation and standards within NASA OSDR.

## Skills

- **Plant molecular biology:** Gateway cloning, Gibson assembly, site-directed mutagenesis, vector construction and design (Vector NTI), transgenic line generation, Arabidopsis genetics and phenotypic analysis
- **Epigenetics & chromatin biology:** histone biology, chromatin remodeling, histone acetylation/methylation analysis, nucleosome organization, ChIP-seq, open-chromatin/accessibility analysis
- **Protein structure & function:** domain-level mutagenesis, intrinsically disordered region (IDR) analysis, protein conformation and structure-function studies
- **Omics & data analysis:** RNA-seq and NicE-seq analysis using SalmonProcessor and OmicsForge, LC-MS proteomics, alternative splicing analysis, sequence alignment (BioEdit), whole-genome data visualization with IGV
- **Protein & stress biology:** ubiquitin–proteasome pathway, E3 ligase complexes, protein stability, cold/drought/thermotolerance and senescence signaling
- **Software/tools:** SalmonProcessor, OmicsForge, IGV, BioEdit, Vector NTI, HOMER, GraphPad Prism, Bio-Rad CFX Maestro (RT-qPCR analysis), confocal imaging analysis (Zeiss ZEN, Leica LAS X), ImageJ

## Publications (SCIE/ESCI)

1 Ali A, Zareen S, Bader ZE, Park J, Khan IU, Park K, Albakri N, **Bae MJ**, Bressan RA, Pardo JM, Yun DJ, Xu ZY. "A GPS2-like protein interacts with HOS15 and HDA6 to form a repressor complex that regulates ABA signaling and drought adaptation in Arabidopsis." *Plant Communications*, 2026.

2 **Bae MJ**, Zareen S, Yun H, Ali A, Bader ZE, Park K, Albakri N, Jung YJ, Yun DJ, Park J. "Unveiling the epigenetic and stress-responsive function of histone H1 in plants." *Plant Signaling & Behavior*, 2026.

3 Albakri N, Bader ZE, Park KS, **Bae MJ**, Zareen S, Yun H, Pradhan S, Yun DJ, Park J. "HOS15 Modulates Alternative Splicing Homeostasis Under Cold Stress in Arabidopsis Thaliana." *Journal of Plant Biology*, 2025.

4 Park K, Lim CJ, Albakri N, **Bae MJ**, Park J. "HOS15 Contributes to Thermotolerance Through Destabilization of HD2C in Arabidopsis." *Journal of Plant Biology*, 2025.

5 Zareen S, Ali A, **Bae MJ**, Albakri N, Park K, Yun H, Yun DJ, Park J. "HOS15 together with PWR–HDA9 positively regulates dark-induced senescence in Arabidopsis." *Plant Signaling & Behavior*, 2025.

6 Bader ZE, **Bae MJ**, Ali A, Park J, Baek D, Yun DJ. "GIGANTEA-ENHANCED EM LEVEL complex initiates drought escape response via dual function of ABA synthesis and flowering promotion." *Plant Signaling & Behavior*, 2023.

7 Lim CJ, Park KS, Ali A, Park J, Ryou SM, Shen M, Khan HA, Bader ZE, Zareen S, **Bae MJ**, Choi JH, Xu ZY, Pardo JM, Yun DJ. "Negative regulation of floral transition in Arabidopsis by HOS15-PWR-HDA9 complex." *Frontiers in Plant Science*, 2023.

8 Lim CJ, Park J, Shen M, Park HJ, Cheong MS, Park K, Baek D, **Bae MJ**, Ali A, Jan M, Lee SY, Lee BH, Kim WY, Pardo JM, Yun DJ. "The Histone-Modifying Complex PWR/HOS15/HD2C Epigenetically Regulates Cold Tolerance." *Plant Physiology*, 2020.

Manuscripts in preparation (targeting PNAS): Mechanisms of Histone H1 degradation under cold stress; HOS15-dimerization promotes HD2C degradation during Cold Stress.

## Patents

- Vector for improving productivity of cellulose and transformed mouseear cress by the same — KR 10-2023-0113597
- Vector for improving productivity and storage capacity of vegetable oil and transformed plant by the same — KR 10-2024-0168464
- Vector for improving productivity of cellulose and transformed mouseear cress by the same — KR 10-2024-0168432
- Sucrose-accumulating plant and method for preparing the same — KR 10-2025-0104961
- Ethanol production and tolerance enhanced yeast strain and method for producing ethanol using the same — KR 10-2025-0104960
- Recombinant Kluyveromyces marxianus strain with engineered xylose metabolic pathway and method for producing ethanol using the same — KR 10-2025-0104960

## Awards and Honors

- HOPE Fellow, HOPE Meetings with Nobel Laureates, Japan Society for the Promotion of Science (JSPS), Japan, Mar. 2024
- Excellent Poster Award, Korean Society of Plant Biologists (KSPB), Korea, Nov. 2021

## Conference Presentations

- 2026 GPSRC International Conference, Konkuk University, Korea, May 2026
- 2026 Winter Conference of the Korean Society of Plant Biologists, Korea University, Korea, Feb. 2026 — Poster
- 2nd Global Plant Stress Research Center (GPSRC) & 2025 KSMCB Plant Molecular Biology International Symposium, Seoul, Korea, Aug. 2025 — Organizer
- 15th HOPE Meeting, Kyoto, Japan, Feb. 2024 — Poster
- 2024 International Conference of the Korean Society of Plant Biologists, Seogwipo-si, Korea, Nov. 2024 — Oral Presentation
- 2022 International Conference of the Korean Society of Plant Biologists, Busan, Korea, Nov. 2022 — Poster
- 2021 International Conference of the Korean Society of Plant Biologists, Online, Nov. 2021 — Poster
- 2020 International Conference of the Korean Society of Plant Biologists, Online, Nov. 2020 — Poster
- 2019 Annual Conference of the Korean Society of Plant Biologists, Buyeo, Korea, Nov. 2019 — Poster
- 2018 International Conference of the Korean Society of Plant Biologists, Busan, Korea, Nov. 2018 — Poster